A Practical Guide to Expectation maximization algorithm

TOPIC -- EXPECTATION MAXIMIZATION ALGORITHM

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Variants A number of methods have been proposed to accelerate the sometimes slow convergence of the EM algorithm, such as those using conjugate gradient and modified Newton's methods (Newton–Raphson). Also, EM can be used with constrained estimation methods. Parameter-expanded expectation maximization (PX-EM) algorithm often provides speed up by "us[ing] a `covariance adjustment' to correct the analysis of the M step, capitalising on extra information captured in the imputed complete data". Expectation conditional maximization (ECM) replaces each M step with a sequence of conditional maximization (CM) steps in which each parameter θi is maximized individually, conditionally on the other parameters remaining fixed. Itself can be extended into the Expectation conditional maximization either (ECME) algorithm. This idea is further extended in generalized expectation maximization (GEM) algorithm, in which is sought only an increase in the objective function F for both the E step and M step as described in the As a maximization–maximization procedure section. GEM is further developed in a distributed environment and shows promising results. It is also possible to consider the EM algorithm as a subclass of the MM (Majorize/Minimize or Minorize/Maximize, depending on context) algorithm, and therefore use any machinery developed in the more general case.

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μ 2 (t + 1) = ∑ i = 1 n T 2 , i (t) x i ∑ i = 1 n T 2 , i (t) , Σ 2 (t + 1) = ∑ i = 1 n T 2 , i (t) (x i − μ 2 (t + 1)) (x i − μ 2 (t + 1)) ■ ∑ i = 1 n T 2 , i (t) . {\displaystyle {\begin{aligned}{\boldsymbol {\mu }}_{2}^{(t+1)}&={\frac {\sum \limits _{i=1}^{n}T_{2,i}^{(t)}\mathbf {x} _{i}}{\sum \limits _{i=1}^{n}T_{2,i}^{(t)}}},\\[1ex]\Sigma _{2}^{(t+1)}&={\frac {\sum \limits _{i=1}^{n}T_{2,i}^{(t)}\left(\mathbf {x} _{i}-{\boldsymbol {\mu }}_{2}^{(t+1)}\right)\left(\mathbf {x} _{i}-{\boldsymbol {\mu }}_{2}^{(t+1)}\right)^{\top }}{\sum \limits _{i=1}^{n}T_{2,i}^{(t)}}}.\end{aligned}}}

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We take the expectation over possible values of the unknown data Z {\displaystyle \mathbf {Z} } under the current parameter estimate θ (t) {\displaystyle \theta ^{(t)}} by multiplying both sides by p (Z ■ X , θ (t)) {\displaystyle p(\mathbf {Z} \mid \mathbf {X} ,{\boldsymbol {\theta }}^{(t)})} and summing (or integrating) over Z {\displaystyle \mathbf {Z} } . The left-hand side is the expectation of a constant, so we get:

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Properties Although an EM iteration does increase the observed data (i.e., marginal) likelihood function, no guarantee exists that the sequence converges to a maximum likelihood estimator. For multimodal distributions, this means that an EM algorithm may converge to a local maximum of the observed data likelihood function, depending on starting values. A variety of heuristic or metaheuristic approaches exist to escape a local maximum, such as random-restart hill climbing (starting with several different random initial estimates ■ θ (t) {\displaystyle {\boldsymbol {\theta }}^{(t)}} ■), or applying simulated annealing methods. EM is especially useful when the likelihood is an exponential family, see Sundberg (2019, Ch. 8) for a comprehensive treatment: the E step becomes the sum of expectations of sufficient statistics, and the M step involves maximizing a linear function. In such a case, it is usually possible to derive closed-form expression updates for each step, using the Sundberg formula (proved and published by Rolf Sundberg, based on unpublished results of Per Martin-Löf and Anders Martin-Löf). The EM method was modified to compute maximum a posteriori (MAP) estimates for Bayesian inference in the original paper by Dempster, Laird, and Rubin. Other methods exist to find maximum likelihood estimates, such as gradient descent, conjugate gradient, or variants of the Gauss–Newton algorithm. Unlike EM, such methods typically require the evaluation of first and/or second derivatives of the likelihood function.

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E-step Operate a Kalman filter or a minimum-variance smoother designed with current parameter estimates to obtain updated state estimates. M-step Use the filtered or smoothed state estimates within maximum-likelihood calculations to obtain updated parameter estimates. Suppose that a Kalman filter or minimum-variance smoother operates on measurements of a single-input-single-output system that possess additive white noise. An updated measurement noise variance estimate can be obtained from the maximum likelihood calculation

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In information geometry, the E step and the M step are interpreted as projections under dual affine connections, called the e-connection and the m-connection; the Kullback–Leibler divergence can also be understood in these terms.

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First, initialize the parameters θ {\displaystyle {\boldsymbol {\theta }}} to some random values. Compute the probability of each possible value of ■ Z {\displaystyle \mathbf {Z} } ■, given ■ θ {\displaystyle {\boldsymbol {\theta }}} ■. Then, use the just-computed values of Z {\displaystyle \mathbf {Z} } to compute a better estimate for the parameters ■ θ {\displaystyle {\boldsymbol {\theta }}} ■. Iterate steps 2 and 3 until convergence. The algorithm as just described monotonically approaches a local minimum of the cost function.

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X i ■ (Z i = 1) ~ N d (μ 1 , Σ 1) , X i ■ (Z i = 2) ~ N d (μ 2 , Σ 2) , {\displaystyle {\begin{aligned}X_{i}\mid (Z_{i}=1)&\sim {\mathcal {N}}_{d}({\boldsymbol {\mu }}_{1},\Sigma _{1}),\\X_{i}\mid (Z_{i}=2)&\sim {\mathcal {N}}_{d}({\boldsymbol {\mu }}_{2},\Sigma _{2}),\end{aligned}}}

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log ■ p (X ■ θ) = log ■ p (X , Z ■ θ) – log ■ p (Z ■ X , θ) . {\displaystyle \log p(\mathbf {X} \mid {\boldsymbol {\theta }})=\log p(\mathbf {X} ,\mathbf {Z} \mid {\boldsymbol {\theta }})-\log p(\mathbf {Z} \mid \mathbf {X} ,{\boldsymbol {\theta }}).}

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F ^ = ∑ k = 1 N (x ^ k + 1 – F ^ x ^ k) 2 ∑ k = 1 N x ^ k 2 . {\displaystyle {\widehat {F}}={\frac {\sum \limits _{k=1}^{N}{\left({\widehat {x}}_{k+1}-{\widehat {F}}{\widehat {x}}_{k}\right)}^{2}}{\sum \limits _{k=1}^{N}{\widehat {x}}_{k}^{2}}}.}

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In words, choosing θ {\displaystyle {\boldsymbol {\theta }}} to improve Q (θ ■ θ (t)) {\displaystyle Q({\boldsymbol {\theta }}\mid {\boldsymbol {\theta }}^{(t)})} causes log ■ p (X ■ θ) {\displaystyle \log p(\mathbf {X} \mid {\boldsymbol {\theta }})} to improve at least as much.